# SpectralBridge
# MESMA readiness audit

Scientific and computational assessment
14 October 2026

**Executive finding:** useful for controlled spectral-library diagnostics, but the current evidence does not support operational ecological abundance estimates.

Distinct spectra do not guarantee identifiable endmembers or recoverable fractions. The bounded held-out benchmarks show substantial convex ambiguity and fraction error. Physical scale/purity/background gaps and absent independent cover truth block a stronger readiness claim.

3,639 polygon identities | 4,807 paired polygon-years | 122 source flights
360 supported scheme/fold tests | 369,048 fitted controlled-mixture results
These simulations and overlapping folds are not independent field replicates.

Guide: the twenty scientific answers come first, followed by material interpretation, selection assumptions, uncertainty, verification and primary references. Six main vector figures and a coverage supplement follow the narrative. Exact memberships, constraints, source hashes and numerical tables are preserved with the report source.

**REAL ABUNDANCE VALIDATION = NOT YET AVAILABLE**

# Executive scientific answer: is the library ready for MESMA?

**Conditionally useful for spectral/mixture diagnostics; NOT operationally validated for material abundances.** Existing spectral separation is not mixture identifiability. Composite endmembers, convex ambiguity, held-out fraction errors, background gaps, unverified per-source scales and absent independent fractions prevent a defensible operational pass. This is a negative/qualified readiness result, not a recommendation to run all spectra through a raster solver.

## 1. Which material classes are supported?

Researcher Deciduous/Evergreen/Herbaceous/Woody shrub/Non-vegetated & dead and five inferred PFTs support labelled-library tests. Rock, Bare and some dead standing/downed classes have explicit subcategory annotations and bounded tests. Those are canopy/background proxies, not verified pure endmembers. MATERIAL_CLASS_AUDIT documents counts and rare/incomplete tags. Green vegetation versus soil/shade material fractions are not yet validated.

## 2. Which distinguishable classes are mixture-ambiguous?

| scheme | sensor | mean class ambiguity rate |
|---|---|---|
| inferred pft | NEON | 0.712 |
| inferred pft | OLI | 0.700 |
| material | NEON | 0.812 |
| material | OLI | 0.794 |
| researcher | NEON | 0.723 |
| researcher | OLI | 0.725 |

These are mean class ambiguity rates in held-out plant folds at relative tolerance 0.01, with correct candidate and other-class convex fits. Class-specific combinations and tolerance sensitivities are retained. Strong historical DBS-ECT/DBT-ECT significance does not eliminate convex overlap; weak DBT-ECS is not treated as an interchangeable material merely because an old test failed.

Selector sensitivity on the SAME plant folds, using at most TWO other materials and a common tolerance of 0.01:

| scheme | sensor | selection | ambiguity rate |
|---|---|---|---|
| inferred pft | NEON | EAR | 0.713 |
| inferred pft | NEON | MASA | 0.708 |
| inferred pft | NEON | all proposed candidates | 0.908 |
| inferred pft | NEON | medoid | 0.664 |
| inferred pft | OLI | EAR | 0.699 |
| inferred pft | OLI | MASA | 0.694 |
| inferred pft | OLI | all proposed candidates | 0.860 |
| inferred pft | OLI | medoid | 0.643 |
| material | NEON | EAR | 0.812 |
| material | NEON | MASA | 0.860 |
| material | NEON | all proposed candidates | 0.906 |

| scheme | sensor | selection | ambiguity rate |
|---|---|---|---|
| material | NEON | medoid | 0.759 |
| material | OLI | EAR | 0.791 |
| material | OLI | MASA | 0.839 |
| material | OLI | all proposed candidates | 0.920 |
| material | OLI | medoid | 0.746 |
| researcher | NEON | EAR | 0.720 |
| researcher | NEON | MASA | 0.736 |
| researcher | NEON | all proposed candidates | 0.835 |
| researcher | NEON | medoid | 0.651 |
| researcher | OLI | EAR | 0.722 |
| researcher | OLI | MASA | 0.735 |
| researcher | OLI | all proposed candidates | 0.826 |
| researcher | OLI | medoid | 0.649 |

More proposed candidates improve own-class coverage but also create more wrong-class convex combinations. This check does not use the test data for selection. It guards against blaming intrinsic material physics for one EAR/shade-aware prototype choice. Candidate budget and model assumptions still limit generalization.

Class-specific inferred-PFT ambiguity (pooled held-out polygon-year records across three plant folds):

| sensor | label | ambiguity rate | median correct class RMSE | median incorrect RMSE |
|---|---|---|---|---|
| NEON | DBS | 0.920 | 0.376 | 0.081 |
| NEON | DBT | 0.948 | 0.719 | 0.092 |
| NEON | EBS | 0.114 | 0.123 | 0.293 |
| NEON | ECS | 0.767 | 0.204 | 0.140 |
| NEON | ECT | 0.809 | 0.270 | 0.183 |
| OLI | DBS | 0.920 | 0.289 | 0.062 |
| OLI | DBT | 0.944 | 0.559 | 0.086 |
| OLI | EBS | 0.089 | 0.097 | 0.245 |
| OLI | ECS | 0.767 | 0.161 | 0.110 |
| OLI | ECT | 0.778 | 0.196 | 0.139 |

## 3. Which classes are inadequately sampled?

Dead - Debris has one identity; dead herbaceous is rare/incompletely annotated; named taxa have severe long tails; some PFTs are species-confounded. Rare/missing class/fold cells stay insufficient, never merged to inflate performance. Some year-transfer directions have no disjoint training support. See original material counts and fold_support.csv.

## 4. Missing backgrounds?

No verified pure soil, litter, water, snow or measured shade classes. Bare is not proof of soil, and shaded polygons are not shade endmembers. Broad Non-vegetated & dead combines disparate materials.

## 5. How many endmembers before a plateau?

Not established. Optimization probes 1,2,5,10 entries within up to 12 proposed training identities/class. Larger 20/50/100 trials are explicitly outside this budget; nearest-neighbor accumulation separately extends up to 500 where support allows. A bounded curve cannot identify a universal optimal count, and adding candidates can increase mixture ambiguity. library_size_fraction_recovery.csv measures this directly on the same mixtures at 1/2/5/10 entries where exact model counts permit.

| scheme | sensor | requested per class | fraction RMSE |
|---|---|---|---|
| inferred pft | NEON | 1 | 0.376 |
| inferred pft | NEON | 2 | 0.344 |
| inferred pft | NEON | 5 | 0.325 |
| inferred pft | NEON | 10 | 0.290 |
| inferred pft | OLI | 1 | 0.360 |
| inferred pft | OLI | 2 | 0.351 |
| inferred pft | OLI | 5 | 0.326 |
| inferred pft | OLI | 10 | 0.294 |
| material | NEON | 1 | 0.304 |
| material | NEON | 2 | 0.292 |
| material | NEON | 5 | 0.273 |
| material | OLI | 1 | 0.302 |
| material | OLI | 2 | 0.291 |
| material | OLI | 5 | 0.279 |
| researcher | NEON | 1 | 0.384 |
| researcher | NEON | 2 | 0.380 |
| researcher | NEON | 5 | 0.310 |
| researcher | NEON | 10 | 0.306 |
| researcher | OLI | 1 | 0.387 |
| researcher | OLI | 2 | 0.380 |
| researcher | OLI | 5 | 0.317 |
| researcher | OLI | 10 | 0.309 |

## 6. Best selection method on held-out data?

| scheme | sensor | method | fraction RMSE |
|---|---|---|---|
| inferred pft | NEON | random MESMA3 | 0.323 |
| inferred pft | OLI | random MESMA3 | 0.324 |
| material | NEON | CoB MESMA3 | 0.281 |
| material | OLI | CoB MESMA3 | 0.282 |
| researcher | NEON | MASA MESMA3 | 0.341 |
| researcher | OLI | diversity MESMA3 | 0.332 |

These are descriptive winners for the declared bounded pool and controlled plant folds, not globally selected operational methods. Three plant folds and one seeded proposed pool do not identify a generally superior selector. Acquisition-stress variants have different targets and are excluded from this ranking. No held-out result was used to choose candidates or change thresholds. IES training kappa need not optimize fractions or retain rare classes.

## 7. Known-fraction recovery?

| scheme | sensor | fraction RMSE | fraction MAE | class set error |
|---|---|---|---|---|
| inferred pft | NEON | 0.344 | 0.215 | 0.888 |
| inferred pft | OLI | 0.351 | 0.222 | 0.867 |
| material | NEON | 0.292 | 0.166 | 0.947 |
| material | OLI | 0.291 | 0.165 | 0.938 |
| researcher | NEON | 0.380 | 0.255 | 0.904 |
| researcher | OLI | 0.380 | 0.252 | 0.878 |

EAR_MESMA3 values are macro means of class/fold fraction RMSE/MAE, in fraction units (0.10 = ten percentage points). Truth consists of algebraic weights on held-out COMPOSITE polygon curves. Synthetic success would not establish actual cover accuracy; the observed errors are already substantial for many labels. Class-specific material results:

| sensor | label | fraction RMSE | fraction MAE |
|---|---|---|---|
| NEON | Bare | 0.235 | 0.105 |
| NEON | Dead - Woody downed | 0.269 | 0.131 |
| NEON | Dead - Woody standing | 0.464 | 0.353 |
| NEON | Deciduous | 0.270 | 0.135 |
| NEON | Evergreen | 0.260 | 0.170 |
| NEON | Herbaceous | 0.276 | 0.148 |
| NEON | Rock | 0.271 | 0.138 |
| NEON | Woody shrub | 0.293 | 0.151 |
| OLI | Bare | 0.223 | 0.098 |
| OLI | Dead - Woody downed | 0.282 | 0.138 |
| OLI | Dead - Woody standing | 0.462 | 0.350 |

| sensor | label | fraction RMSE | fraction MAE |
|---|---|---|---|
| OLI | Deciduous | 0.264 | 0.125 |
| OLI | Evergreen | 0.265 | 0.174 |
| OLI | Herbaceous | 0.272 | 0.141 |
| OLI | Rock | 0.270 | 0.140 |
| OLI | Woody shrub | 0.290 | 0.153 |

## 8. False positives and false negatives?

| scheme | sensor | macro FPR true zero | macro FNR true 0.2 | macro FNR true 0.5 | macro FNR true 1 |
|---|---|---|---|---|---|
| inferred pft | NEON | 0.320 | 0.561 | 0.473 | 0.383 |
| inferred pft | OLI | 0.324 | 0.626 | 0.458 | 0.383 |
| material | NEON | 0.230 | 0.654 | 0.531 | 0.452 |
| material | OLI | 0.223 | 0.671 | 0.576 | 0.435 |
| researcher | NEON | 0.286 | 0.599 | 0.531 | 0.500 |
| researcher | OLI | 0.277 | 0.557 | 0.561 | 0.517 |

Values are macro class/fold conditional frequencies on controlled mixtures, not field error probabilities.

Detection rule estimated fraction >=0.10 was declared before testing; 0.05/0.20 sensitivities are included. detection_errors.parquet reports per-fraction false-positive rates at true zero and false-negative rates at positive truth, with exact simulation denominators. They depend strongly on hierarchy, component and fraction; no single biological FPR/FNR is justified. Reused source parents invalidate treating all gradient points as independent field samples.

## 9. Reliable abundance detection thresholds?

No defensible population 50/80/95% threshold. Figure 4 shows conditional curves. Three plant folds and one parent set per composition provide too little independent replication, and the sources are not verified pure materials. A nominal detection at one simulation fraction is not a calibrated ecological detection limit.

## 10. NEON to OLI loss?

The table above and paired sensor_comparison.csv use the same parents, fractions and candidate identities. Interpret OLI-minus-NEON fraction RMSE within exact class/fold pairs, not the previous 15/16 biological-test retention statistic. Both sensors can be ambiguous; little additional spectral loss does not mean either recovers fractions well. Weighting differs by declared spectral support. This is synthetic reduction, not independent satellite abundance validation.

## 11. Does MESMA outperform FCLS?

method_comparison.csv compares EAR-limited two/three-component variable candidates with fixed class-medoid FCLS on identical mixtures. Performance is hierarchy-dependent. Extra spectral flexibility does not consistently guarantee lower fraction error, and minimal spectral residual is not a fraction-accuracy selector. No universal MESMA advantage is established.

| scheme | sensor | EAR MESMA3 | FCLS medoid | MESMA3 minus FCLS | MC mean FCLS | MC minus FCLS |
|---|---|---|---|---|---|---|
| inferred pft | NEON | 0.344 | 0.366 | -0.022 | 0.281 | -0.085 |

| scheme | sensor | EAR MESMA3 | FCLS medoid | MESMA3 minus FCLS | MC mean FCLS | MC minus FCLS |
|---|---|---|---|---|---|---|
| inferred pft | OLI | 0.351 | 0.364 | -0.013 | 0.277 | -0.088 |
| material | NEON | 0.292 | 0.295 | -0.003 | 0.233 | -0.062 |
| material | OLI | 0.291 | 0.295 | -0.004 | 0.234 | -0.061 |
| researcher | NEON | 0.380 | 0.342 | 0.037 | 0.279 | -0.063 |
| researcher | OLI | 0.380 | 0.345 | 0.035 | 0.279 | -0.066 |

## 12. Does Monte Carlo help?

MC_mean_FCLS is evaluated on the same sources; uncertainty tables show its fraction errors, empirical width and conditional truth coverage. An ensemble can stabilize some estimates, but empirical 95% intervals are not assumed calibrated. Twelve endmember draws cannot justify tail-probability claims or resolve unobserved materials. The ensemble improves macro fraction RMSE in this benchmark but often increases thresholded class-set error by distributing fractions across absent classes; it is not an unqualified improvement in material detection.

## 13-15. Flight, year and site transfer?

TRANSFERABILITY.md and transferability_summary.csv give class/fold error distributions, support and exact source memberships. All actual flight holdouts were considered; rare/absent class cases remain explicit. Actual flight spectra are used as targets, and repeated IDs are purged globally. Year/site shifts alter class/taxon composition and candidate availability; no isolated environmental penalty is inferred. Repeated NIWO years never count as geographic replication. Unsupported reverse-year directions do not become optimistic estimates.

## 16. Correct material absent?

| scheme | sensor | known false alarm | unknown flag rate |
|---|---|---|---|
| inferred pft | NEON | 0.693 | 0.401 |
| inferred pft | OLI | 0.661 | 0.362 |
| material | NEON | 0.721 | 0.421 |
| material | OLI | 0.727 | 0.449 |
| researcher | NEON | 0.836 | 0.657 |
| researcher | OLI | 0.802 | 0.623 |

These are pure nominal-label missing-class diagnostics against training-only thresholds. Wrong classes can fit well; acceptance does not establish presence. unknown_model_status.parquet restores source year from the exactly matching pure-target ledger and explicitly marks NO_ADEQUATE_LIBRARY_MODEL or LOW_RESIDUAL_NOT_VALIDATED_MEMBERSHIP, not a forced operational class. The rates are descriptive repeated-record frequencies, not calibrated field probabilities. Partly unknown mixtures and new background types need additional validation.

## 17. Detect unreliable estimates?

unreliable_fraction_flags.csv reports the fraction of poor estimates hiding below three residual tolerances and sensitivity of a near-model fraction-range flag. Small residual is insufficient. Model-choice ranges and empirical bundles help expose instability but miss within-composition/unsampled variability. No validated operational rejection algorithm is claimed.

## 18. Applications supported now?

Stored-library QC, labelled material/PFT variability, conditional same-library spectral comparisons, controlled linear-mixture geometry and paired spectral-resolution diagnostics. These are exploratory analyses, not calibrated abundance products.

## 19. Applications not supported?

Real vegetation/background/species cover maps, calibrated uncertainty/detection limits, universal cross-site/year fraction transfer, unlabelled-landscape completeness, real Landsat abundance maps and subtle abundance change detection. Independent REAL ABUNDANCE VALIDATION = NOT YET AVAILABLE.

## 20. Exact work before operational MESMA?

Recover per-flight metadata and processing provenance; audit purity and dated labels; reconcile backgrounds; add missing pure materials and rare independent identities; obtain complete co-registered independent fractions; validate larger nested libraries and source-parent replication; calibrate uncertainty/abstention; separate spectral from actual spatial-support effects and validate real sensors across independent acquisitions. No global coefficient or BRDF/SRF change was made to improve this audit.

## Defensible manuscript statement

"In identity-disjoint controlled mixtures constructed from the current 3,639-identity SpectralBridge library, the tested candidate/model sets showed substantial other-class convex overlap. With two EAR-selected candidates per class and at most three nonnegative, sum-one material components, macro class/fold fraction RMSE was 0.344 for inferred PFTs and 0.380 for researcher categories in corrected NEON space, versus 0.351 and 0.380 in paired synthetic OLI space. Larger candidate sets and alternative selectors changed performance but did not establish operational abundance readiness. These simulations use empirical composite polygon curves; unverified source-scale/purity information and unavailable independent fractional-cover truth preclude a claim of field abundance accuracy."

## Final operational readiness matrix

| material | NEON | Landsat like |
|---|---|---|
| Green vegetation | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Deciduous | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Evergreen | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Herbaceous | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Woody shrub | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| NPV/dead | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Rock | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Bare | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Soil | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Water | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Snow | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Measured shade | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |
| Species-level | INSUFFICIENT EVIDENCE | INSUFFICIENT EVIDENCE |

Each cell lacks independently validated material fractions and source/purity completeness. This does not mean all spectral comparisons are impossible. It means the current evidence does not justify operational ecological abundance claims. Ten separate dimensions are recorded in readiness_dimensions.csv; there is no opaque combined score.

# Material class audit

The library has 3,639 GlobalIDs. A polygon identity is not necessarily an individual living plant, a pure canopy or a physically pure material. Species-null polygons remain through researcher labels. Preserve species, name-derived PFT and researcher categories as different annotation systems.

| label | unique GlobalIDs | sites |
|---|---|---|
| Deciduous | 442 | 3 |
| Evergreen | 1363 | 3 |
| Herbaceous | 677 | 3 |
| Non-vegetated & dead | 414 | 3 |
| Woody shrub | 743 | 3 |

## Existing physical hints, not invented classes

| field | label | unique GlobalIDs | sites |
|---|---|---|---|
| cover subcategory | Bare | 44 | 3 |
| cover subcategory | Dead | 165 | 3 |
| cover subcategory | Forb | 51 | 3 |
| cover subcategory | Grass | 160 | 3 |
| cover subcategory | Mixed Dead / Bare / Rock | 71 | 3 |
| cover subcategory | Mixed grass & forb | 465 | 3 |
| cover subcategory | Rock | 119 | 3 |
| cover subcategory | Woody shrub - Broadleaf | 547 | 3 |
| cover subcategory | Woody shrub - Needleleaf | 48 | 2 |
| cover subcategory | (not recorded) | 1969 | 3 |
| dead subcategory | Dead - Debris | 1 | 1 |
| dead subcategory | Dead - Herbaceous | 18 | 2 |
| dead subcategory | Dead - Woody downed | 62 | 3 |
| dead subcategory | Dead - Woody standing | 119 | 3 |
| dead subcategory | (not recorded) | 3439 | 3 |

The name-derived PFT experiment covers woody taxa: all 677 Herbaceous and all 414 Non-vegetated & dead identities are UNASSIGNED in that crosswalk. It therefore cannot substitute for a full material library. Some assigned PFTs span researcher cover categories; pft_by_cover_category.csv records the exact cross-tabulation.

The broad Non-vegetated & dead category is physically heterogeneous and cannot be treated as a single clean soil/NPV endmember. Rock (119 identities) and Bare (44) exist as explicit labels. Bare is a substrate annotation, not verified mineral soil. Dead standing/downed woody annotations exist; 71 polygons explicitly say Mixed Dead / Bare / Rock. No fraction or purity measurement resolves those mixtures. A Bare record also has a dead-herbaceous subtype: conflicting detail needs review. The five inferred PFT labels are trait annotations, not independent material spectra.

The controlled material experiment conservatively uses concordant category/subcategory labels. Generic Dead records use their recorded dead_subcategory. Fifteen records with missing cover_subcategory but recorded dead subtypes are kept in the inventory and outside this conservative material benchmark pending annotation reconciliation; they are not declared physically unknown or silently merged. Rare Dead - Herbaceous records in the concordant subset fail some/all support gates. Original metadata and fold-support tables expose these choices. The combined researcher category is tested separately to show consequences of a broad hierarchy, not adopted as an operational material.

Measured water, snow, litter, pure shade and verified soil are absent as distinct validated classes. Dead - Debris has one annotated identity and does not establish a litter library. The shaded field describes 206 shaded, 53 mixed, 1,872 sunlit and 1,508 unspecified polygons; it is not a measured black-shade spectrum. Do not rename canopy shadow as a new material or use zero as empirical shade truth.

Recommended hierarchy: retain broad vegetation types as labelled canopy proxies; separate annotated Rock, Bare, woody standing/downed and dead herbaceous where supported; retain mixed/uncertain annotations explicitly. Establish independently measured purity and background fractions before ecological material abundance mapping. Species-level readouts require their own sampling/validation and cannot inherit a PFT result.

# Spectral integrity and variability

Reuse P120's verified compact pairing: 85,706 extracted pixel rows, 12,504 plant/flight observations, 4,807 polygon-years, 3,639 identities, three source grids and 122 flights. The 1,140-column union contains structural NaNs; native records each have 426 finite channels. Existing 5-nm harmonization never bridges gaps >8 nm or absorption intervals 1340-1450 and 1790-2000 nm. The analysis uses the finite intersection of that already validated support and all seven admitted OLI channels; exact coordinates are in executed_design.json. Do not interpolate atmospheric gaps or interpret union-schema missingness as random noise.

Native raw range: 0.0 to 23645.0; corrected range: 0.0 to 621788.625, in STORED UNITS. There are 0 negative and 1,031,605 zero corrected native entries. These counts include all native bands; absorption-supported roughness is separate. Zero is not automatically nodata. No -9999 exact sentinel was observed. Saturation cannot be established from a reflectance number without original source QA/headers.

The merged Parquet has no Arrow unit/scale metadata and historical source headers point to an unavailable archive. An unrelated R10C H5 does not prove scales for these flights. Therefore physical range >1, detector saturation, source-level brightness comparability and absolute reflectance/noise tolerances are BLOCKED. No scale was inferred from observed magnitude. A common training-only RMS divisor stabilizes numerical mixture fits and leaves convex fractions unchanged under one common positive factor; it does not verify per-flight scaling or physical reflectance.

Within-polygon native variance is reduced from all pixels in bounded 2,048-row batches using centered sufficient moments, outside absorption bands and within the inherited supported wavelength range. Pixels are immediately discarded; no persistent pixel cache is created. Population variances require >1 observed pixel for informative within-polygon variability; a zero from one point is not evidence of homogeneity. Tables retain parent count and original source/flight identities. Raw/corrected roughness and extremes are traceable per flight. Large correction extremes may be processing/illumination artifacts; unusual spectra were not deleted or automatically

called ecological variation.

Repeated-observation dispersion partitions in hierarchical_dispersion.csv are separate descriptive views, not additive causal components. Identity, species, class, site, year and acquisition are confounded. The observed between-identity partition is not a genetic variance estimate; within-identity differences include acquisition, environment, year and correction effects. No model was forced to estimate unidentifiable random effects. Figure 1 shows full-value PCA on symlog axes plus class median/IQR traces with real gaps.

# Endmember selection and computational envelope

Training-only selectors are stratified random, Euclidean medoid, EAR, MASA, marginal in-CoB with EAR ties, add/remove training-kappa IES, medoid-seeded farthest-point diversity and a named IES-plus-class-completion hybrid. The last is a bounded adaptation, not a claim to reproduce Roth's full injection schedule. Exact source identities, candidate scores and selected memberships are archived. in/out CoB counts are recomputed in EAR_MASA_CoB.csv from the same training pools and fixed 0.02 relative-RMS threshold. EAR is candidate-plus-ideal-black-shade reconstruction, not raw Euclidean distance; MASA is angular. Self is excluded from EAR/MASA averages. The workflow evaluates these definitions in the declared wavelength-quadrature feature space: NEON EAR/CoB use weighted RMSE and MASA a weighted angle. This is an explicit metric adaptation, not an exact equal-band reproduction of another study. CoB's incremental new coverage differs from a simple ranking of total counts.

All selection is repeated within each fold and evaluated on absent GlobalIDs. IES optimizes training classification kappa; it may omit rare classes and cannot be presumed optimal for abundance. The cap is a global k x class count, not k guaranteed examples in every class. EAR optimizes candidate-plus-shade reconstruction; a no-shade mixture objective can therefore prefer different prototypes. This objective mismatch is not evidence of intrinsic material indistinguishability. ambiguity_selection_sensitivity.csv compares EAR, medoid, MASA and the entire proposed pool with a common two-other-material limit, so conclusions do not rely solely on an EAR choice. Random/medoid/EAR/MASA/diversity and CoB have their stated per-class size constraints; CoB can stop earlier. The hybrid completes omitted classes from minimum-EAR candidates. A method winning here is only a winner for this bounded candidate pool, split design and nominal-label simulations.

Size curves evaluate 1,2,5,10 candidates/class on held-out class reconstruction, including all available lesser sizes for rare classes. library_size_fraction_recovery.csv additionally evaluates those counts on identical held-out synthetic mixtures using EAR where exhaustive models fit the 20,000-model cap; over-budget cases are reported rather than pruned. Requested sizes 20,50,100 are recorded outside the candidate budget. Fraction recovery primarily uses two selected examples/class; additional complexity compares up to two versus three materials. These experiments do NOT establish a globally optimal endmember count or a plateau for full operational libraries. A Pareto decision must jointly consider class coverage, fraction error, ambiguity, model count and runtime, not choose smallest residual alone.

Exact class-distinct model count with C classes and k candidates/class is sum_{r=1..m} choose(C,r) k^r. For C=5, k=100, m=3 this already gives 10,100,500 models; for C=8 it gives 56,280,800. The implemented cap is 20,000, checked before allocating models. Exhaustive convex search is exact within the declared candidate/model set, with all lower-dimensional simplex faces and bounded model chunks. It is verified against an independent constrained optimizer on small random cases. Numerically singular interiors are skipped at an affine Gram eigenvalue cutoff max(1e-12, 1e-10 times the largest eigenvalue); their lower faces remain eligible. Exactness is relative to this declared numerical-rank tolerance. The near-collinear regression example has negligible spectral error but radically different near-equivalent fractions, so it is not declared precisely identifiable. Exact fixed-library FCLS is capped at 12 vertices. No huge source/pixel cache or uncontrolled raster MESMA was launched. Runtime tables name kernel timing separately from I/O/selection/outer overhead; source-dependent timings are evidence, not package constants.

# Controlled mixtures and recovery

Every mixture source is an empirical held-out polygon/year spectrum or, for flight transfer, an actual held-out-flight polygon spectrum. All observations of its GlobalID are purged from training. A labelled polygon is not a validated pure material: known fractions here are algebraic weights on those composite source curves. Even perfect recovery would not validate real ecological cover.

Pairs use 0,.05,.1,.2,.3,.4,.5,.6,.7,.8,.9,.95,1; triples use equal thirds and .1/.2/.7, .05/.45/.5, .8/.1/.1. One source combination per composition/fold is drawn with a fixed seed. Reusing parents across gradient points makes simulations dependent. Source IDs include year; synthetic_mixture_truth.parquet is separate from candidate memberships. No candidate spectrum generates its own test mixture.

Primary methods are physically constrained classical variable-endmember search (two/three material limits), fixed class-medoid FCLS and empirical Monte Carlo one-candidate-per-class FCLS. All estimated fractions are nonnegative and sum to one. Boundaries are solved as lower-dimensional faces, not unconstrained fractions clipped and renormalized. No sparse L1 benchmark is claimed: L1 is constant on the nonnegative sum-one simplex. A nonconstant, validated sparse alternative is deferred.

Fraction bias, MAE/RMSE, false positives/negatives, selected class-set error, spectral RMSE/angle and sum/negative violations are archived. Class-set error thresholds BOTH true and estimated fractions at 0.10; its estimand differs from detection false negatives, which treat ANY positive true fraction as present. Neither is called field classification accuracy. MC does not enumerate MESMA class sets: its near_compositions=0 records no enumerated sets, not a claim of unique composition. Detection is predeclared at estimated fraction >=0.10, with .05/.20 sensitivity. False positives have true fraction zero; false negatives have positive truth but fail detection. Conditional detection frequencies carry explicit denominators. No 50/80/95% population detection threshold is defensible from three plant folds and one parent draw per composition. The curves are diagnostics, not calibrated operating characteristics.

Acquisition stress draws signed differences between repeated TRAINING identities' year spectra at amplitudes 0.25 and 0.5, coupled across sensors. This includes biological/environmental/processing variability and is NOT instrument noise. Folds without eligible repeats do not receive fabricated noise. A general real-Landsat residual distribution is not imposed on other sites. Ideal black-shade sensitivity injects 25% zero spectrum into an FCLS model; the zero is hypothetical and cannot calibrate actual canopy shade. shade_formulation_comparison.csv compares no-shade, explicit black shade and shade-normalized estimates with matched total-footprint or non-shade-composition truth. Shade normalization changes the estimand and does not validate real shade or total cover.

# Fraction uncertainty, shade and unknowns

| scheme | sensor | mean empirical interval width | conditional truth coverage | mean fraction RMSE |
|---|---|---|---|---|
| inferred pft | NEON | 0.595 | 0.867 | 0.281 |
| inferred pft | OLI | 0.594 | 0.865 | 0.277 |
| material | NEON | 0.388 | 0.863 | 0.233 |
| material | OLI | 0.390 | 0.868 | 0.234 |
| researcher | NEON | 0.587 | 0.842 | 0.279 |
| researcher | OLI | 0.598 | 0.869 | 0.279 |

Twelve empirical random one-candidate-per-class FCLS draws produce an ensemble mean and empirical 2.5/97.5 percentiles. These are ENDMember-choice stability ranges, not posterior distributions or confidence intervals. Twelve draws are sparse for tail estimation. The nominal 95% reference is evaluated against controlled synthetic truth, not asserted calibrated. Finite candidate pools omit unobserved variability; the parent spectra themselves may be mixed. No independent flight-based sensor-noise model is available.

The best-per-composition near-model range is a lower bound on model ambiguity, not a full fraction interval. Small spectral residuals can coexist with large fraction RMSE; unreliable_fraction_flags.csv quantifies false-confidence sensitivities without choosing a threshold after seeing results. Thresholds 0.005/0.01/0.02 are relative spectral tolerances and fraction error 0.10/0.20 are illustrative decision aids, not universal pass criteria. Do not operationally accept an unknown merely because its residual is small.

Removing an entire class tests unknown-material masquerading at pure nominal-label endpoints. More realistic unknown mixtures, novel species/backgrounds, complete-source-image OOD validation and calibrated abstention remain required. Independent empirical shade spectra are missing. A 25% idealized zero-shade simulation probes algebra only; shade-normalized fractions describe conditional non-shade composition, while total fractions describe the footprint. Nonlinearity cannot be diagnosed uniquely from these residuals: endmember incompleteness, purity, atmosphere/correction and variable scaling are competing explanations. No nonlinear model was implemented to explain away failure.

# Validation, scientific gates and reproducibility

Full configured repository suite: **612 passed, 8 skipped, 29 warnings** (final log in checks/). Fifteen focused regression tests also pass. All ordinary tests are offline and use small analytic/fixture data, not production coefficients, scene count constants or live STAC.

The exact convex solver was checked against an independent SLSQP constrained optimizer on random small cases and against analytically obvious perfect mixtures, vertices, boundaries, explicit idealized shade and a distinct class that is exactly a mixture of others. Tests verify nonnegative/sum-one fractions, true active faces, no fake extra composition from zero fractions, near-collinear fraction ambiguity, model-budget refusal, EAR versus simple Euclidean distance, deterministic CoB/IES/MASA/random/diversity selections, grouped identity leakage, coupled sensors, nested JSON flight histories and actual-flight target replacement. Class-size limits are explicit; no test establishes scientific field accuracy.

The complete eligible library enters metadata/QC/coverage audits. The controlled source/algorithm first pass preceded expansion. Final expanded outputs record 360 supported scheme/fold tests and 369,048 fitted mixture estimates, with 1,282,485 explicit fold-membership records. Unsupported class/fold/year directions stay in the ledger. All available actual-flight fold masks are checked as decoded source flights, not history strings. Every simulated parent is in test membership and absent from ALL training acquisitions. Selection and candidate scoring are training-only. Both representations use the same biological source IDs, fraction vectors and selected candidate IDs. Additional size/shade and selector-ambiguity sensitivities retain those splits and thresholds.

The interpreter verifies all 3,642 owned analysis output checksums and exact package implementation hashes, source SHA256, grouped memberships and zero fraction-constraint violations. Large CSV exports are represented as compressed Parquet with the original text-table checksum, row count and lossless numerical export ledger. They are not copies of the source pixel library. Original pixel data are read in bounded batches only for native QC and immediately reduced to compact moments. Existing synthetic PFT/sensor/real-Landsat results are not overwritten.

Source SHA256: cd3a8afd2697f09d1f4749f92df8a5e2cc71ee12490d38de12f21f90b2a75978. Canonical source size is 255,546,633 bytes. Exact source/configuration/code identities and fold checkpoint ledgers support restart. Final no-change restart and evidence/file checks are recorded in CHECKS.md.

Endmember-budget timing is source dependent and not a package constant.

Scientific gates are NOT passing tests: merged-library physical scale/offset/source QA and endmember purity remain unverified; material backgrounds are incomplete; full same-site source imagery/complete cover maps are absent; independent reference fractions are unavailable. R10C n=1 real Landsat agreement is not a noise model or fraction validation for these sites. Therefore REAL ABUNDANCE VALIDATION = NOT YET AVAILABLE. Shade is idealized, noise is acquisition variability stress, class-set error is a thresholded simulation metric, and Monte Carlo intervals are uncalibrated stability ranges. No synthetic mixture success or failed conditional benchmark is called a universal ecological result.

# Primary definitions and implementation decisions

Roberts et al. (1998), https://doi.org/10.1016/S0034-4257(98)00037-6 : MESMA varies spectral candidates and model size per observation and evaluates physical fractions plus residual criteria. Their numerical thresholds are study-specific; importing 0.025 as a universal pass threshold is unjustified. More components can increase competing admissible models rather than improve material discrimination.

Dennison & Roberts (2003), https://doi.org/10.1016/S0034-4257(03)00135-4 ; author preprint https://content.csbs.utah.edu/~pdennison/pubs/rse_dennison2003a_preprint.pdf : EAR averages each candidate's modelling RMSE over other same-class spectra, excluding itself. Here the explicit two-endmember model is candidate plus zero photometric shade, with non-shade fraction constrained [0,1]. This stricter fraction convention is disclosed; no hidden brightness normalization.

Roth, Dennison & Roberts (2012), https://doi.org/10.1016/j.rse.2012.08.030 ; https://content.csbs.utah.edu/~pdennison/reprints/denn/2012_Roth_etal_RSE.pdf : MASA ranks same-class mean angles; CoB greedily covers previously unmodelled same-class training spectra using an RMSE threshold, breaking ties by EAR. IES starts with the best training-kappa pair and adds/removes candidates while kappa improves. It need not retain rare classes. Size limits/candidate budgets and class-completion hybrids are named adaptations, not exact reproductions of every paper setting. The source's equations and sections 2.4, 3.4 were checked before implementation.

Schaaf et al. (2011), https://doi.org/10.2747/1548-1603.48.3.324 ; author version https://content.csbs.utah.edu/~pdennison/reprints/denn/2011_Schaaf_etal_GISRS.pdf supports the kappa-based IES add/remove definition; this training selection objective must be evaluated on separately held-out identities.

Somers, Asner, Tits & Coppin (2011), https://doi.org/10.1016/j.rse.2011.03.003 ; author/institutional record https://experts.azregents.edu/en/publications/endmember-variability-in-spectral-mixture-analysis-a-review/ : endmember variability includes spatial/temporal/material effects; multiple candidate spectra, weighting/feature choices and physical models address different parts of that problem. An empirical Monte Carlo bundle is variability sensitivity, not automatic uncertainty calibration.

FCLS is nonnegative least squares on the sum-one simplex. The implementation will independently verify small-model exact active-face projection against constrained numerical optimization. L1 on that simplex is constant: it cannot be represented as a meaningful sparse benchmark without changing the model or penalty. Spectral angle is shape-only and must accompany magnitude/fraction errors. Absolute reflectance thresholds remain blocked by the merged library's missing per-source unit headers.

# Scientific figure appendix

01 library atlas

02 identifiability

03 library optimization

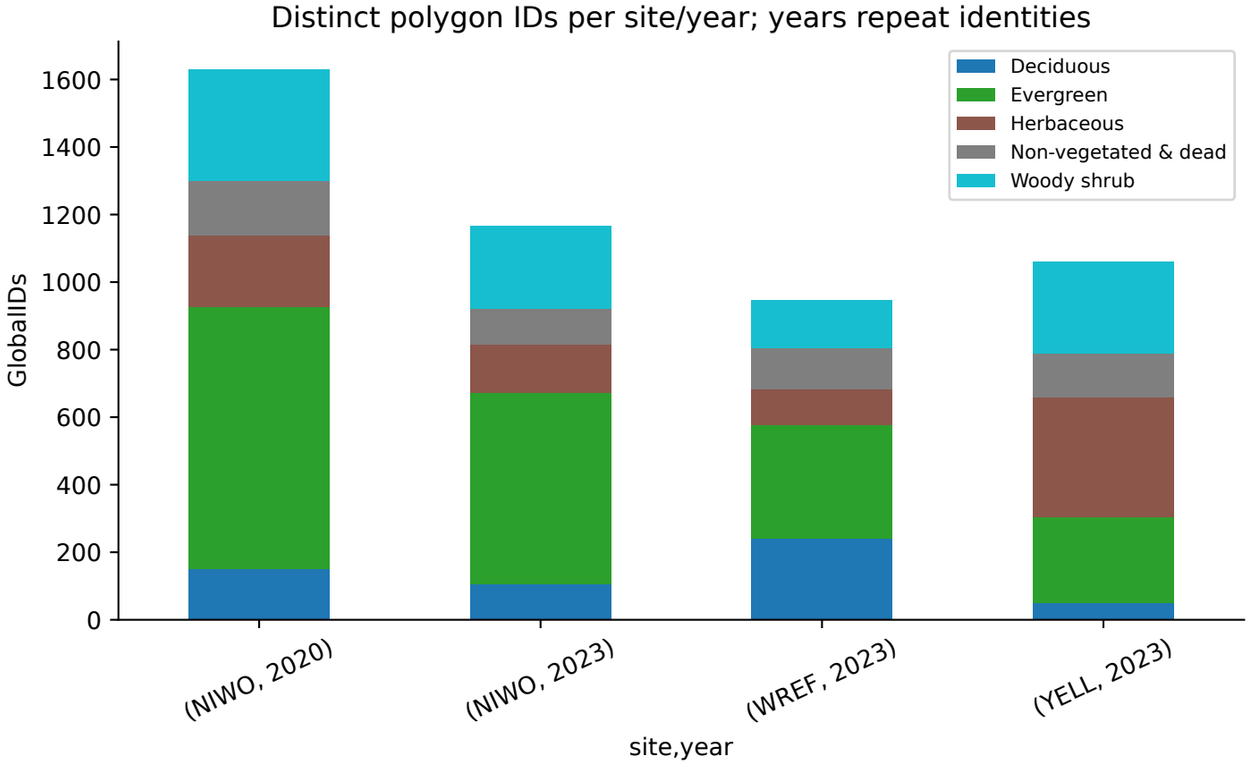
04 fraction recovery

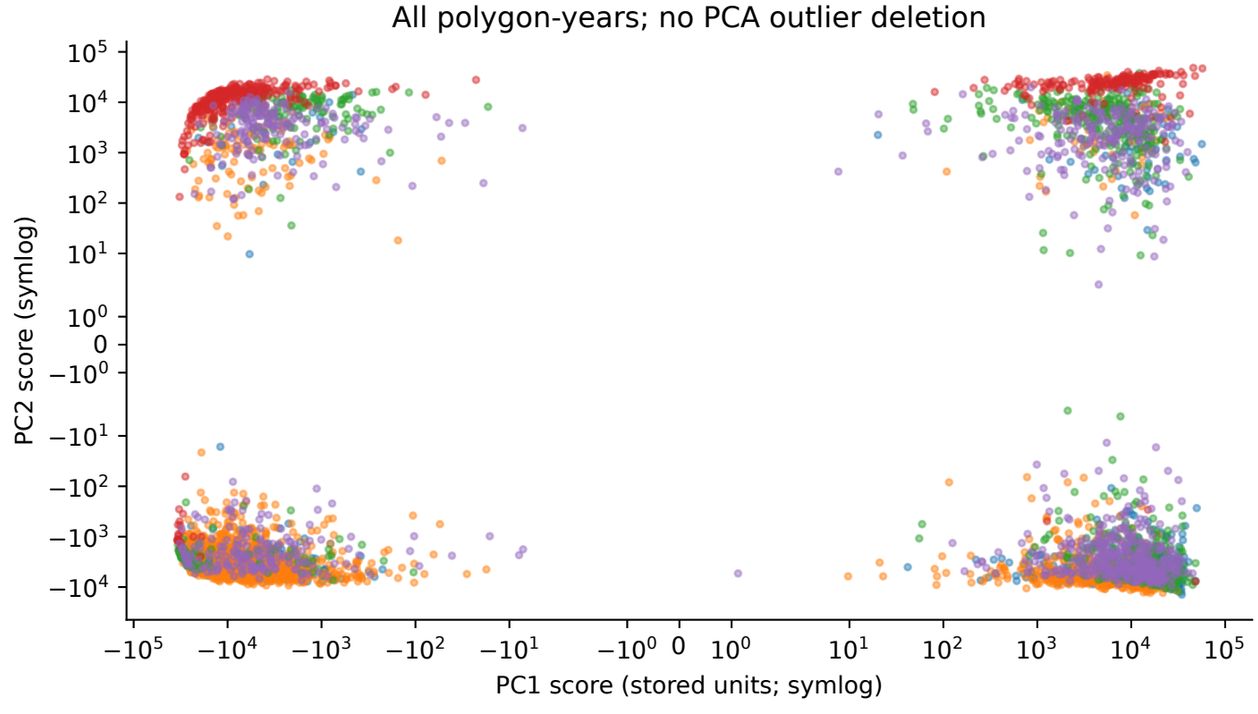
05 sensor and transfer

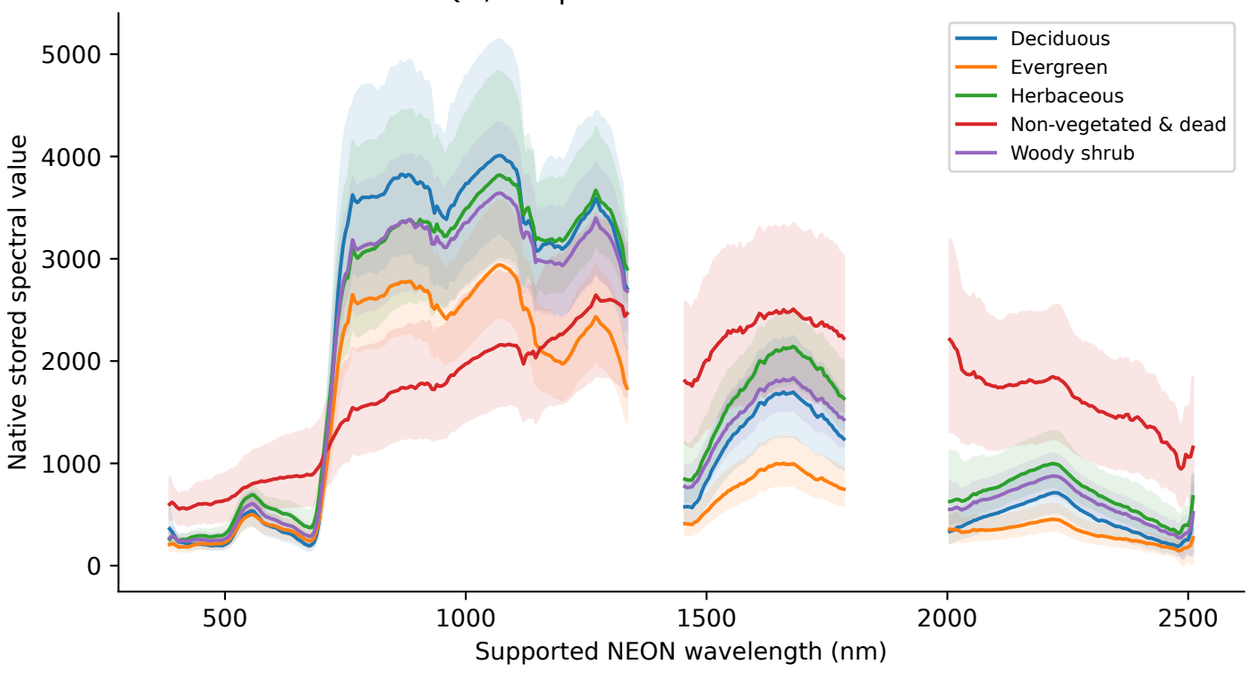
06 readiness dimensions

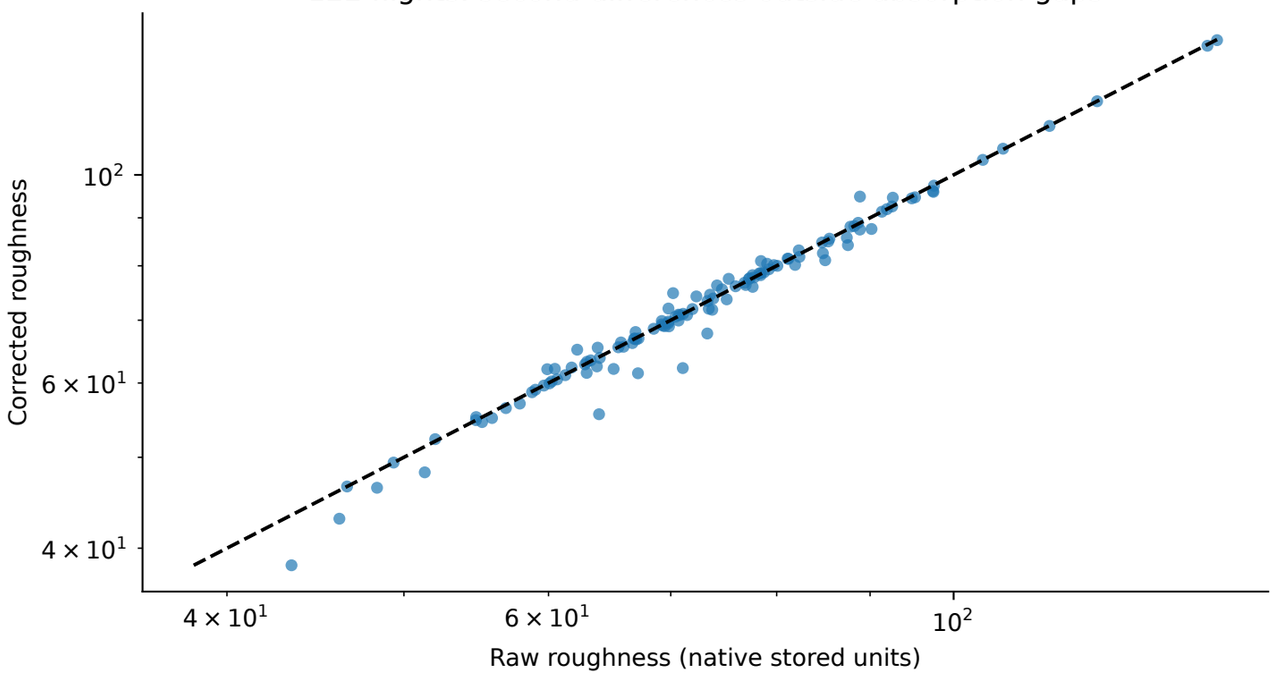
S1 group coverage

Figures retain original vector dimensions for readable zooming. Six primary families plus one coverage supplement. Native units are explicitly unverified; no displayed residual tolerance is presented as a universal reflectance threshold.

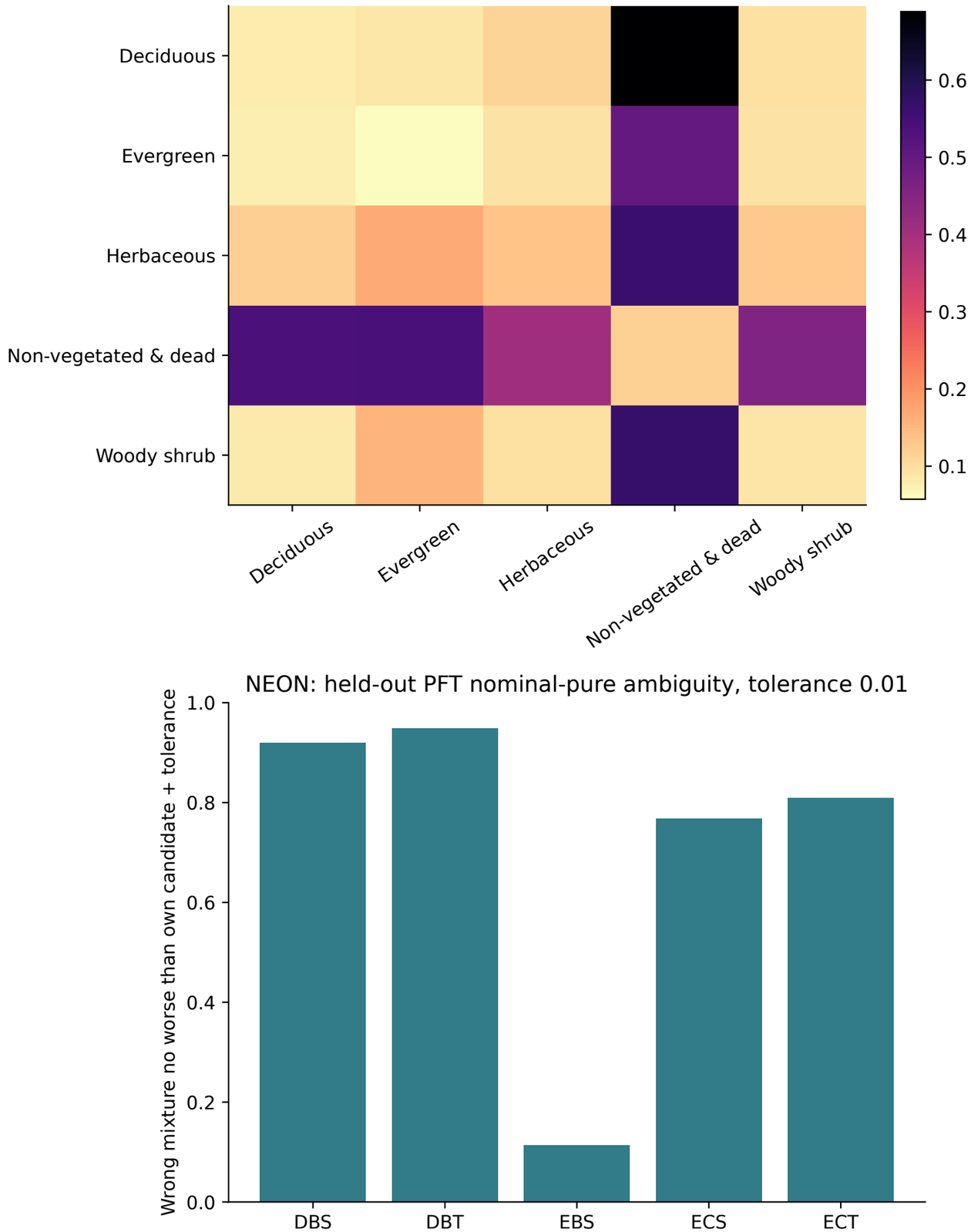


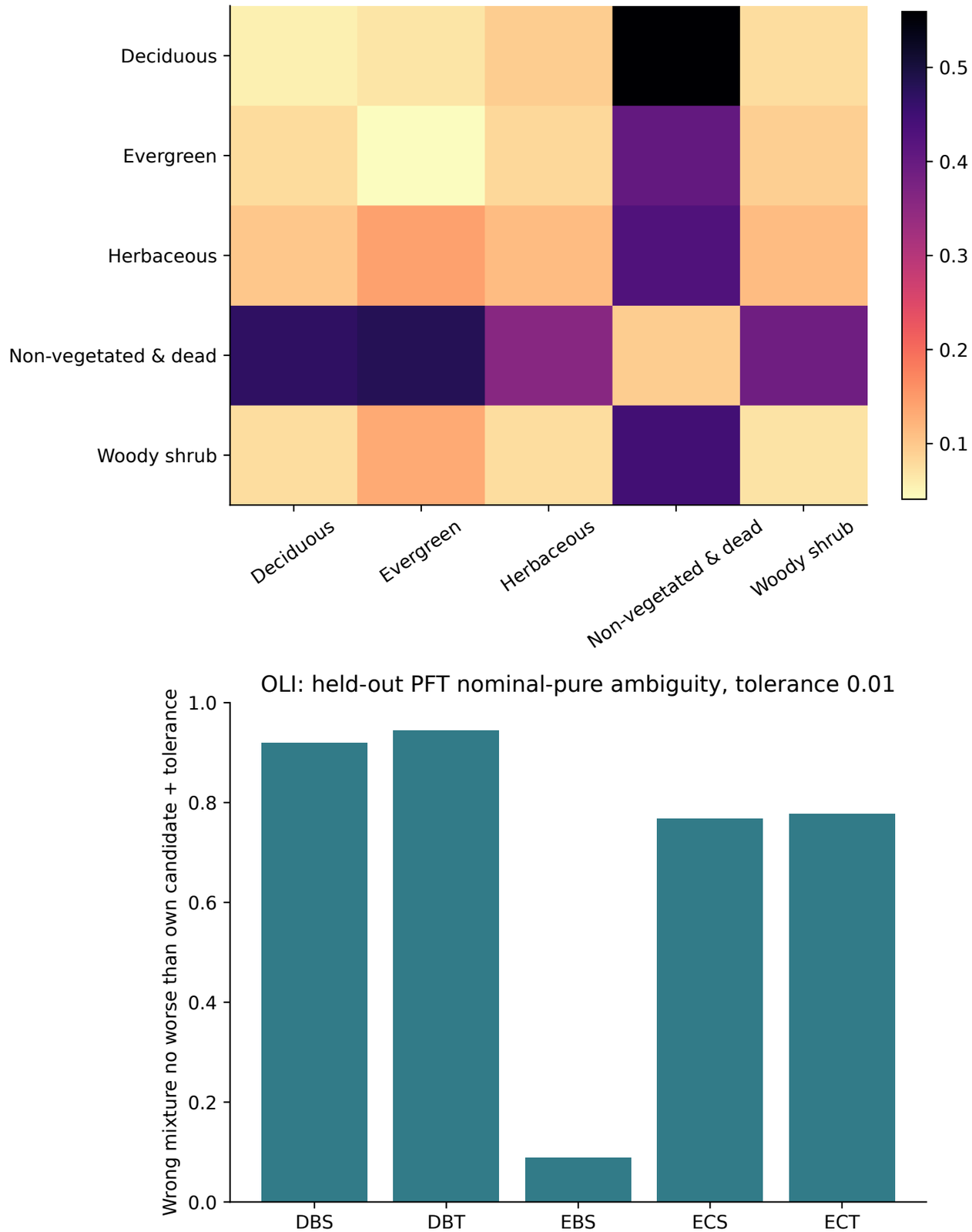




Polygon-year and flight summaries are descriptive, spatially/repeatedly sampled. PCA includes all values; symlog axes disclose the large dynamic range.
Physical reflectance scaling, saturation flags and purity are not verified for this merged library. Gaps are not interpolated.



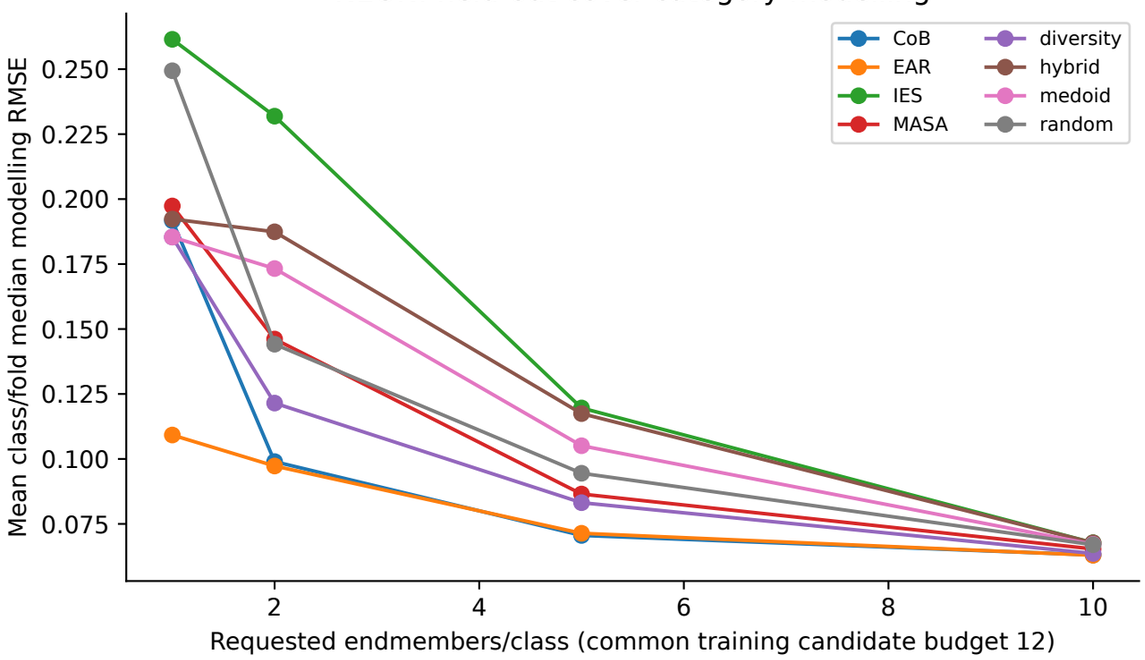




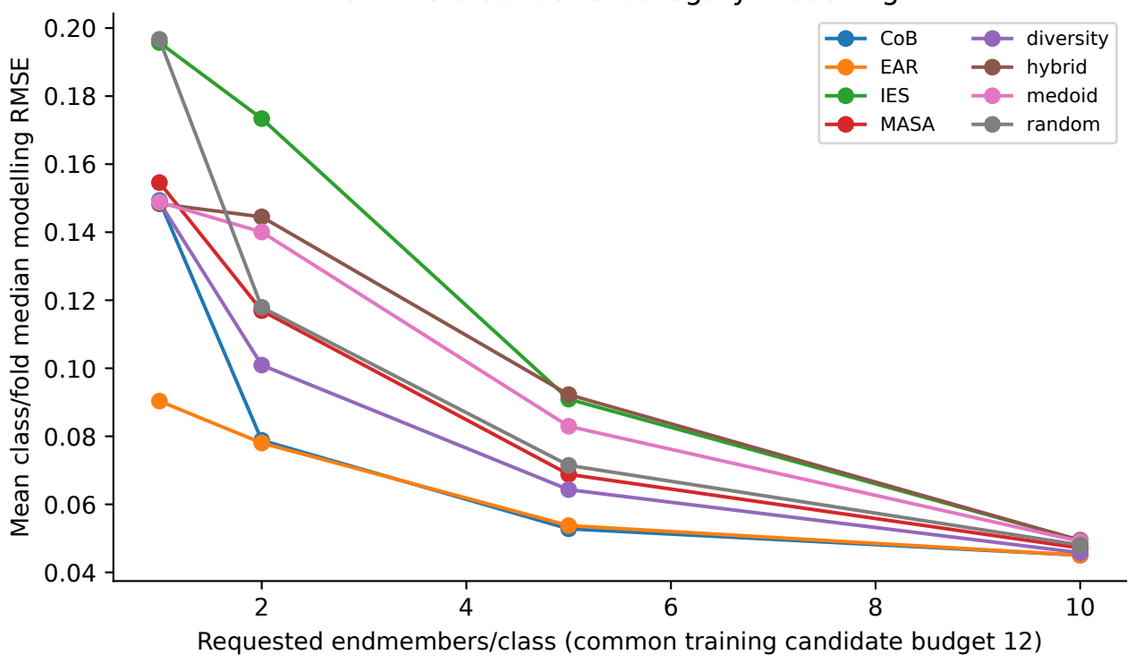



Top panels use the declared candidate+ideal black shade model (EAR-style), and are directed. Bottom uses no-shade candidate vs convex OTHER-class mixtures.
Tolerance is 0.01 of a training-only RMS scale, not reflectance or a universal noise floor. A biological spectral difference is not a fraction-identifiability result.

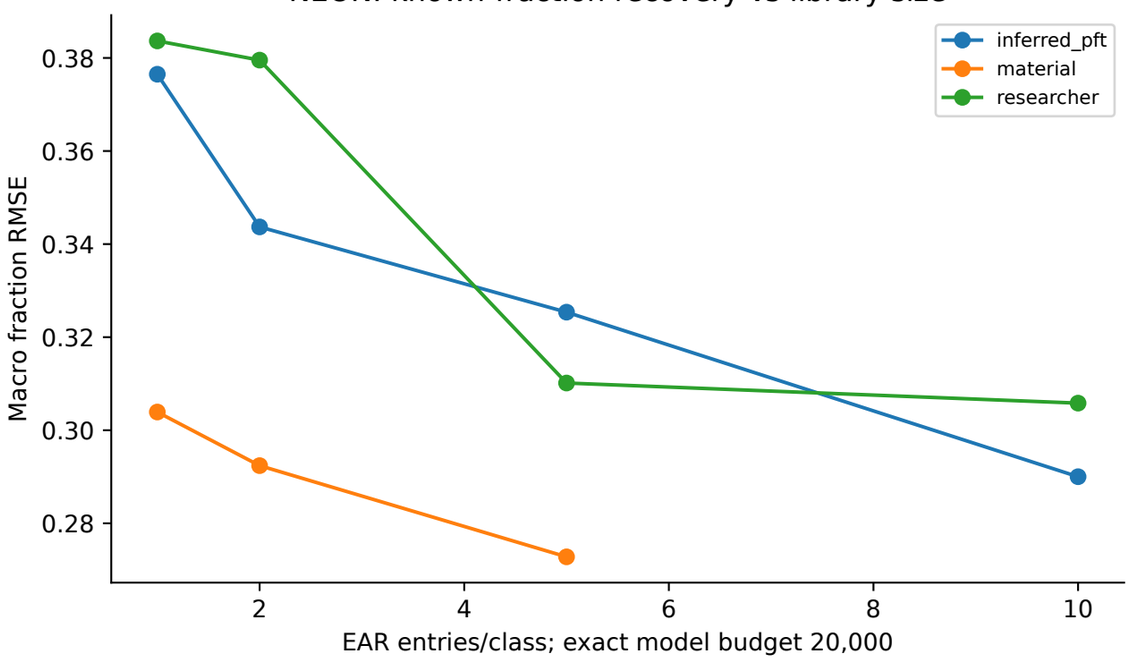
NEON: held-out cover-category modelling



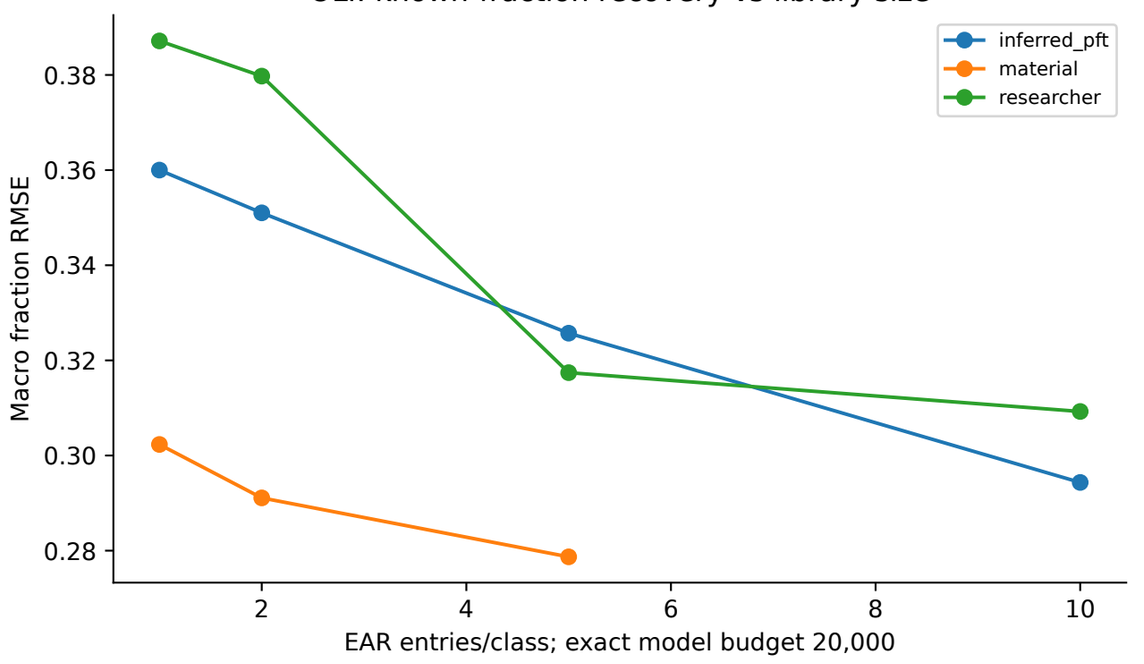

OLI: held-out cover-category modelling



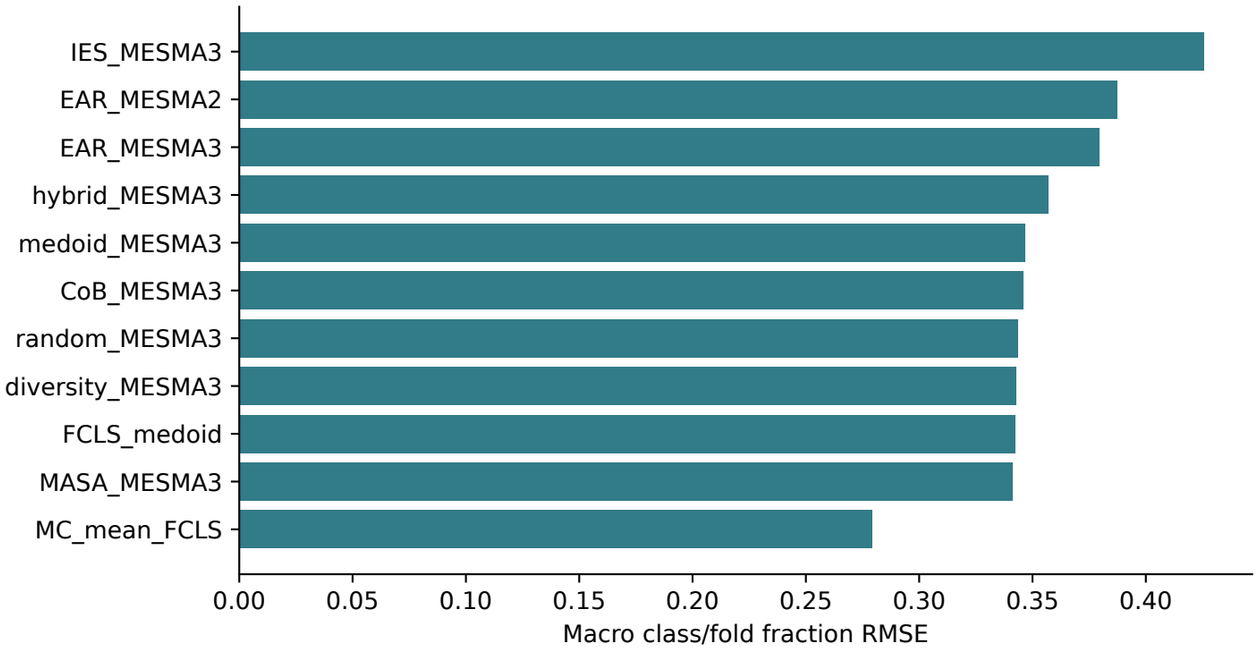

NEON: known-fraction recovery vs library size



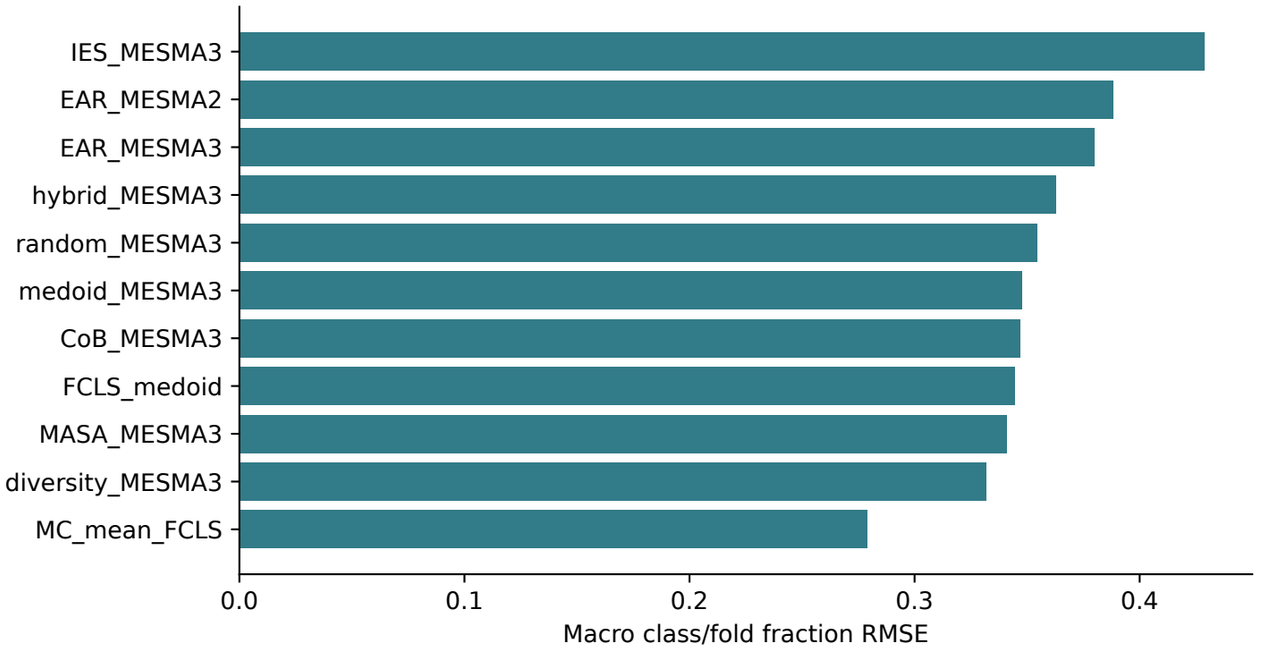

OLI: known-fraction recovery vs library size




NEON: bounded controlled recovery; no method selected on test




OLI: bounded controlled recovery; no method selected on test




EAR/MASA multi-entry ranks and size-limited IES are explicit adaptations. CoB can stop early; IES may omit a rare class. Exact selections are archived.
12 candidate identities/class is a controlled budget, not evidence of saturation at 12. Requested 20/50/100 entries remain untested; accumulation coverage is separately measured up to 500.

DBS vs ECT

DBT vs ECT

DBT vs ECS

NEON
OLI

Estimated fraction, median and IQR

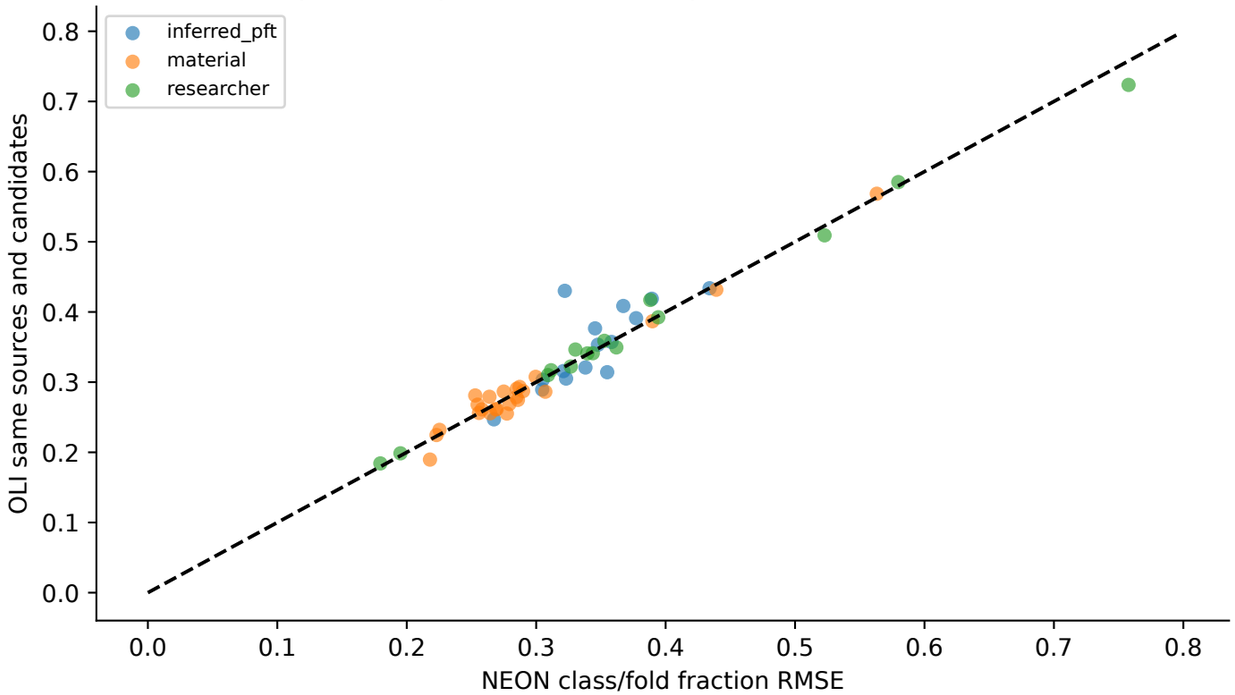
True DBS fraction

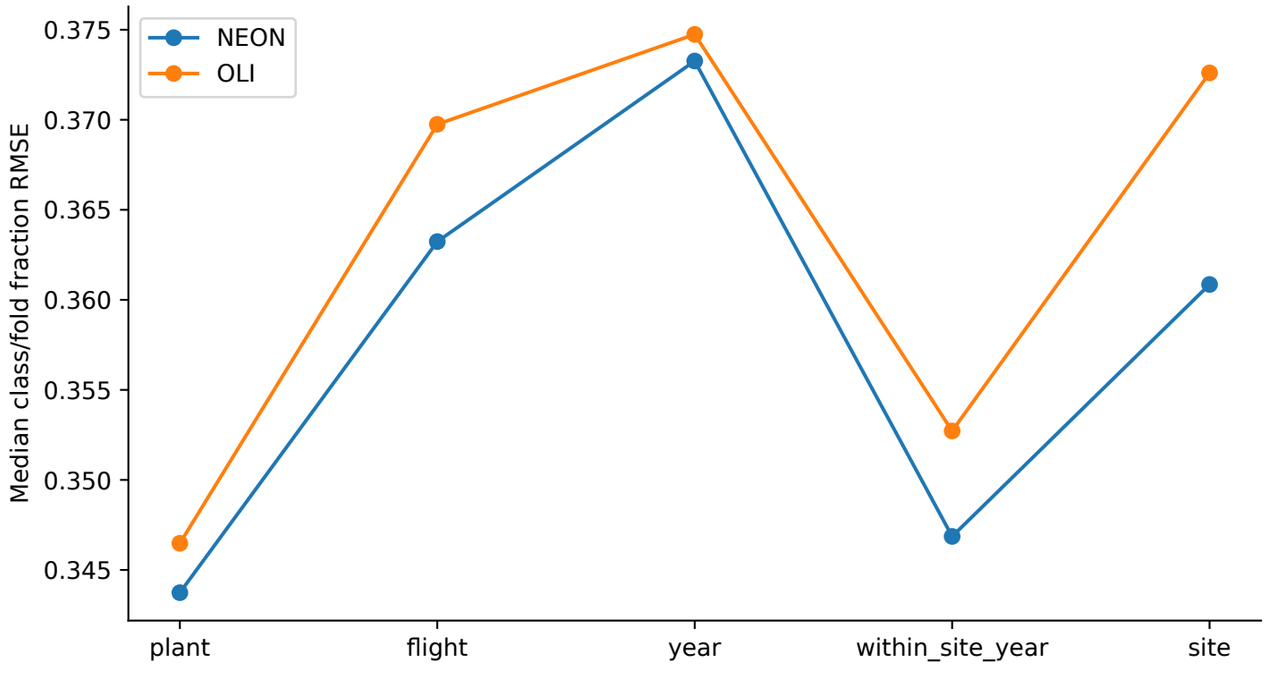
True DBT fraction

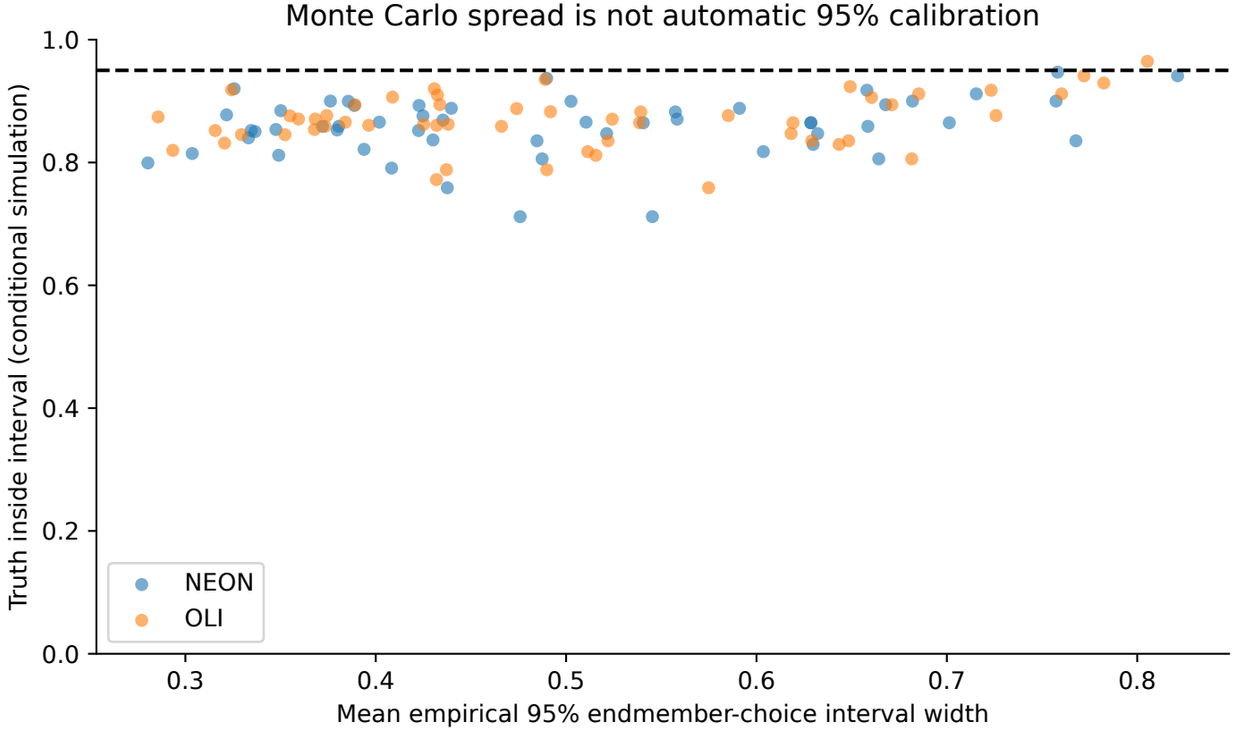
At true zero, frequency is false-positive rate

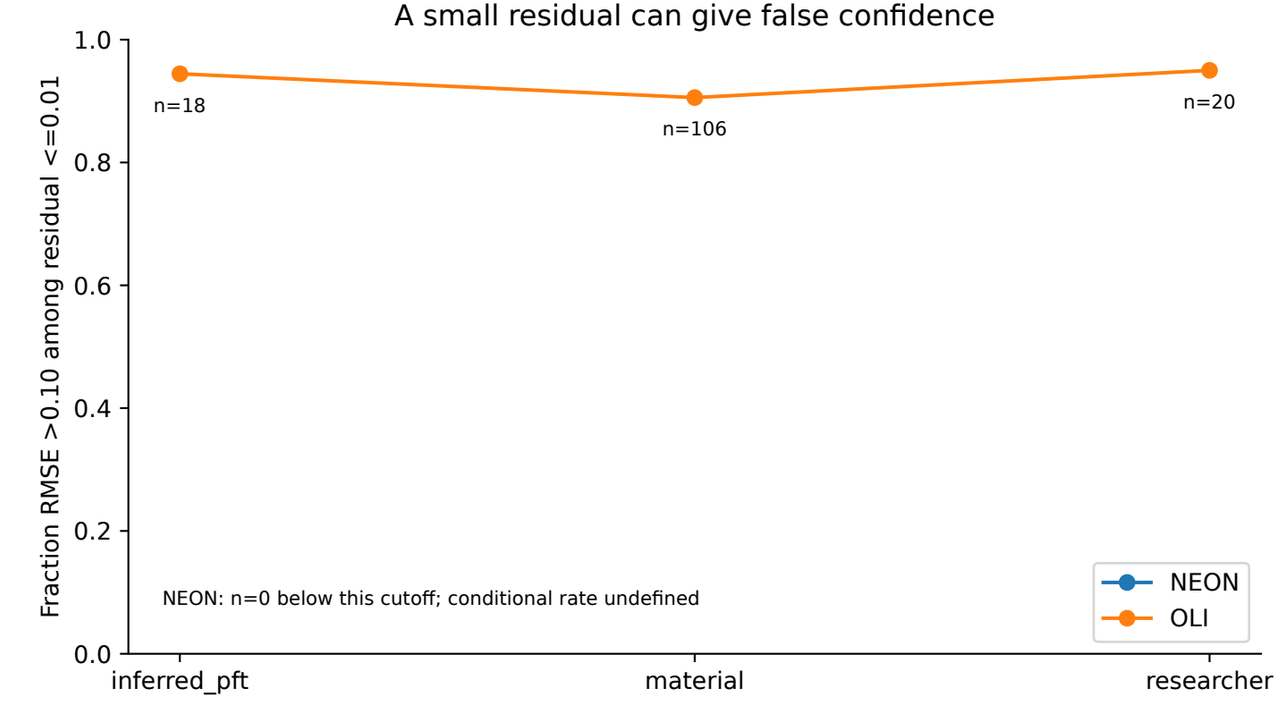
Conditional detection frequency (estimate >= 0.10)

Mixtures are generated from held-out empirical polygon spectra, absent from the candidate pool in ALL acquisitions. Identical source IDs/fractions across sensors.
Curves share parents; displayed counts are simulation draws, not independent biological replicates. These are nominal-label mixtures, not validated pure-material fractional cover.




NEON uses disconnected wavelength quadrature, OLI equal channels; differences include the declared spectral weighting. No per-spectrum brightness normalization.
All flight targets use actual held-out-flight spectra. Repeated identities are purged across dates; missing class/fold support remains in fold_support.csv. Real abundance truth is unavailable.

### Spectral integrity

PARTIAL: grids/masks valid; absolute scales missing

### Material coverage

PARTIAL: Rock/Bare/dead labels; water/snow/litter/shade gaps

### Representativeness

PARTIAL: library holdouts, no same-site full imagery

### Distinguishability

CONDITIONAL: sampled-group differences previously shown

### Mixture identifiability

TESTED: substantial other-class convex ambiguity

### Fraction recovery

TESTED: conditional synthetic fraction errors, no real truth

### Transferability

TESTED: bounded grouped holdouts; some unavailable

### Sensor robustness

TESTED: paired synthetic OLI only; no abundance ground truth

### Uncertainty calibration

NOT ESTABLISHED: empirical bundles not calibrated intervals

### Computation

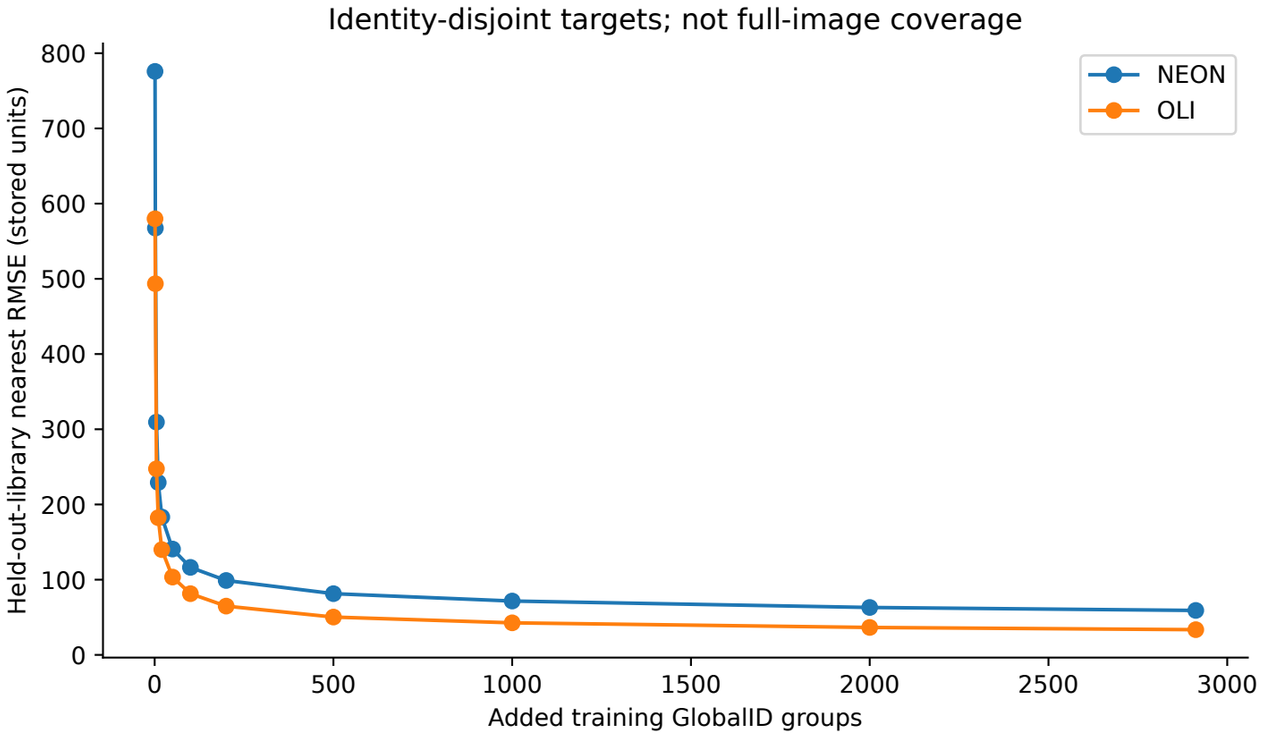
FEASIBLE: bounded candidates, exact search and checkpoints

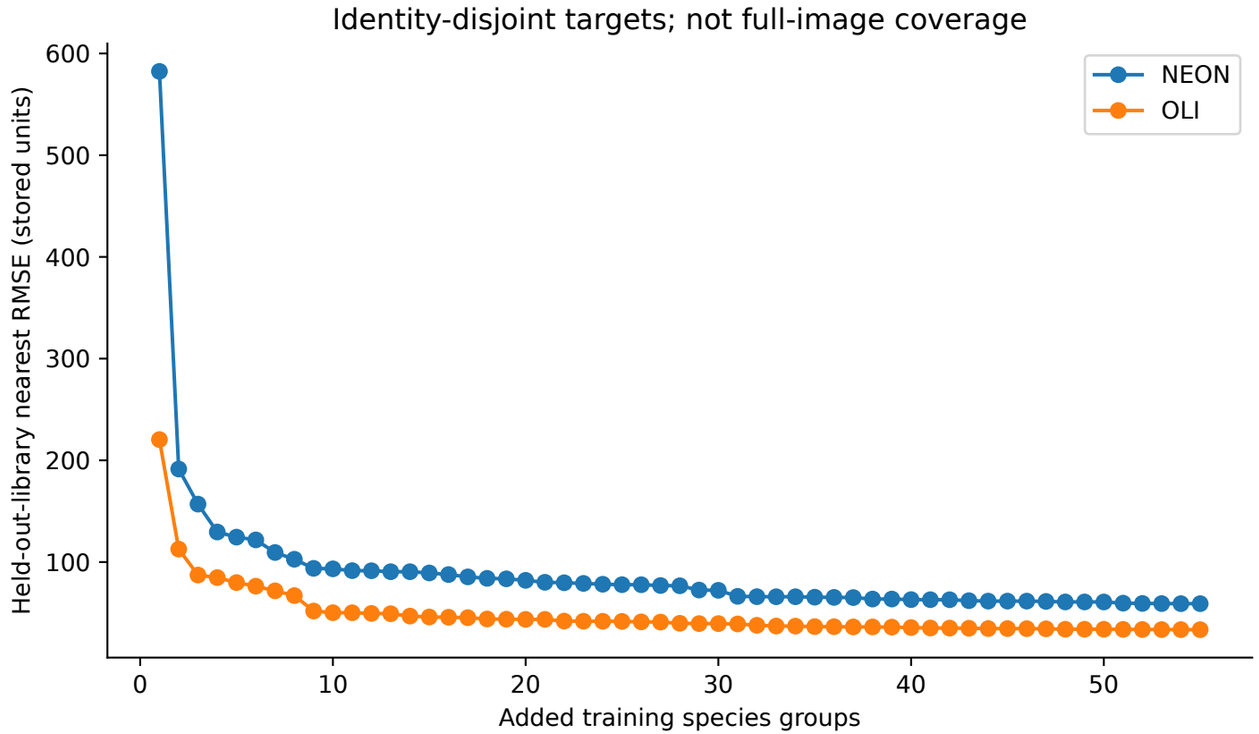
## Operational material-abundance readiness

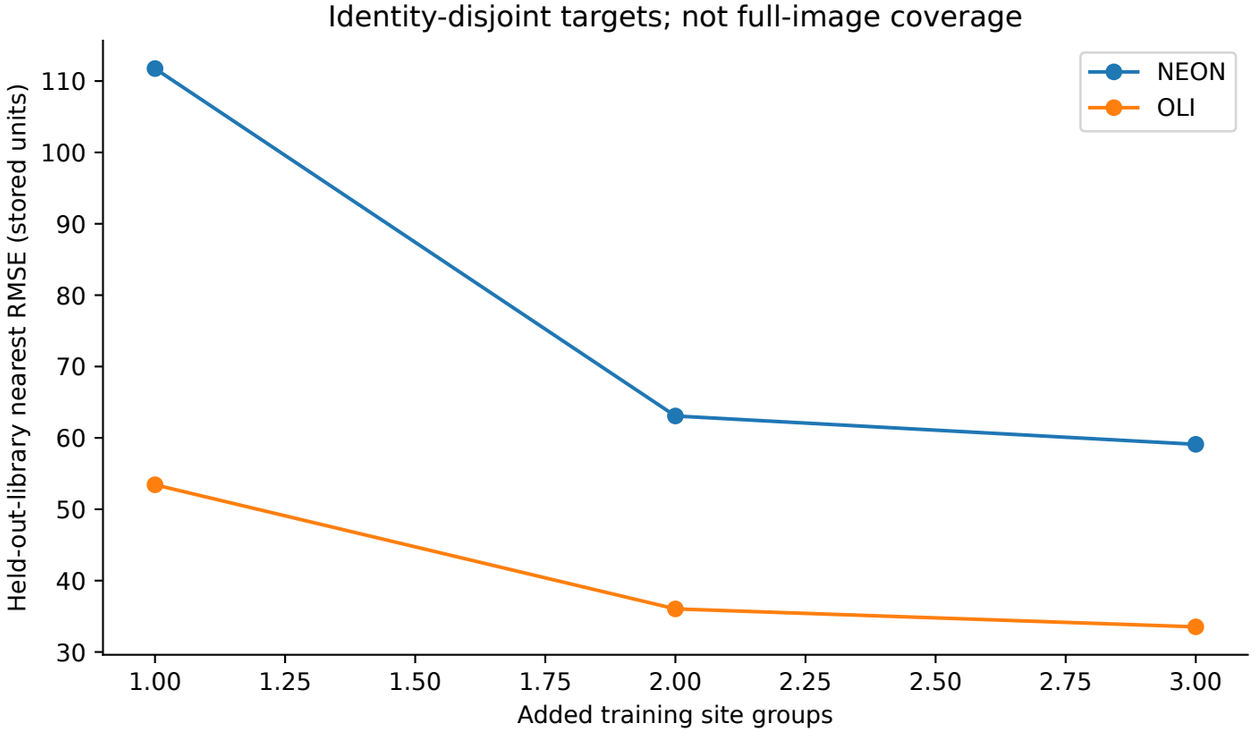
| | |
|---|---|
| Green vegetation | INSUFFICIENT EVIDENCE |
| Deciduous | INSUFFICIENT EVIDENCE |
| Evergreen | INSUFFICIENT EVIDENCE |
| Herbaceous | INSUFFICIENT EVIDENCE |
| Woody shrub | INSUFFICIENT EVIDENCE |
| NPV / dead | INSUFFICIENT EVIDENCE |
| Rock | INSUFFICIENT EVIDENCE |
| Bare (not verified soil) | INSUFFICIENT EVIDENCE |
| Soil | INSUFFICIENT EVIDENCE |
| Water / snow | INSUFFICIENT EVIDENCE |
| Measured shade | INSUFFICIENT EVIDENCE |
| Species-level | INSUFFICIENT EVIDENCE |

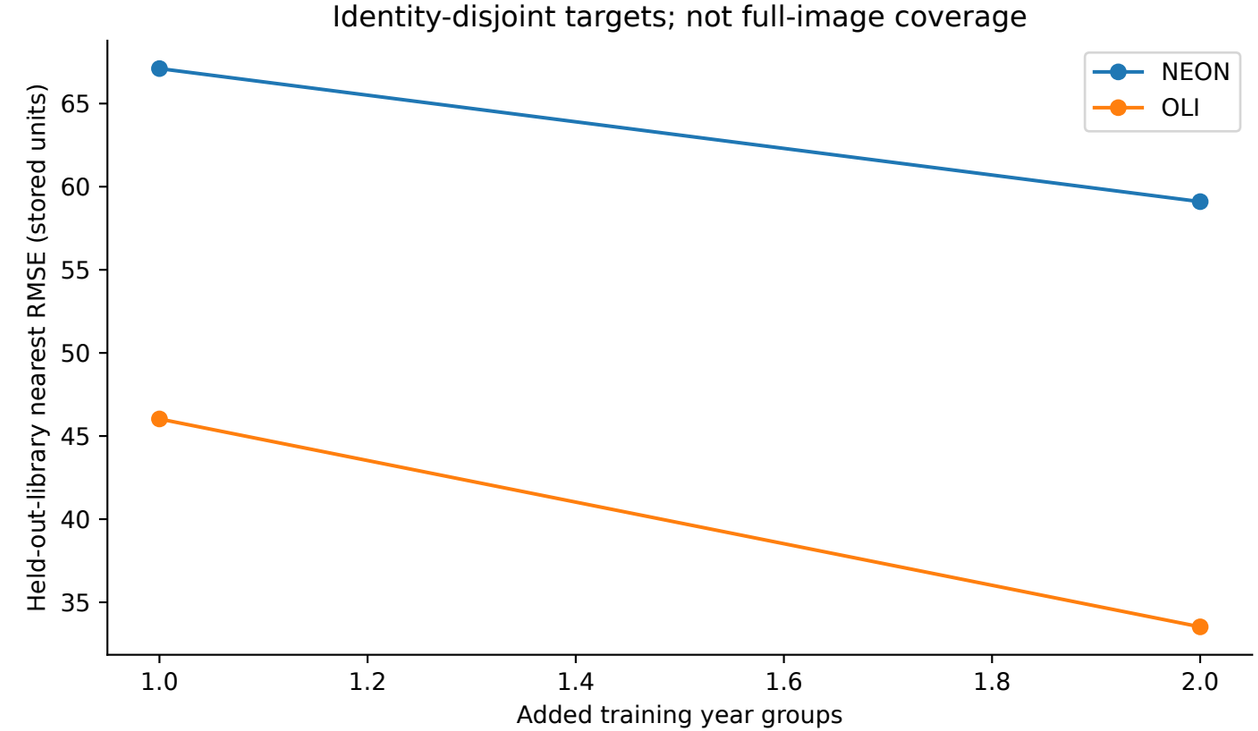
Same status for NEON and Landsat-like abundance.
No cell is promoted to READY from a synthetic-only fit.

No opaque readiness score. Current support is for labelled-library diagnostics and explicitly conditional synthetic mixtures, not operational ecological abundance.
Missing source-scale/purity/reference fractions are blocking data problems. Each material conclusion is tied to quantitative tables or explicitly insufficient evidence.




Ten group-ordering replicates; medians shown. These reflect coverage of held-out library records, not landscape completeness or independent ecological replication.
Group additions change class composition/environment together; no causal site/year effect is identified.